

Brian Ngo

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EDUCATION

Master of Science: Computer Science

University of the Pacific - Stockton, CA

Aug 2023 - Expected May 2025

GPA: 4.0

Relevant Coursework:

- Advanced Software Engineering
- Computational Complexity
- Virtual Reality
- Modern Backend Software Development
- Game Theory
- Brain Machine Interface

Bachelor of Science: Computer Science

University of the Pacific - Stockton, CA

Aug 2021 - Dec 2024

GPA: 3.94

Relevant Coursework:

- Programming Languages
- Computer Systems
- Algorithms
- Discrete Math
- Data Structures
- Application Development
- Operating Systems
- Digital Design
- Computing Theory
- Computer Game Technologies
- Engineering Ethics
- Computer Simulation
- Computer Networking

WORK EXPERIENCE

Pacific Technology - University of the Pacific

Stockton, CA

Enterprise Application Developer 1

Jan 2025 - Present

- Collaborated in a team of nine to convert a total of 258 COBOL programs to Python
- Developed reusable Python libraries, reducing code redundancy by over 500 lines
- Leveraged the SCRUM framework to ensure optimal business outcomes

Networking Technician

Jan 2023 - Dec 2024

- Worked in a team of five to provide layer 2 and 3 support for students and staff on campus
- Initiated an automation pipeline in Python that reduced report processing time by 80%
- Assisted in implementing an optimized wireless deployment, resulting in a -57 dBm average signal strength
- Diagnosed and reported critical issues Aruba CPPM inner radius server, improving reliability

Technical Support

Aug 2021 - Dec 2022

- Partnered with eight others to provide layer 1 and 2 support for staff and students on campus
- Resolved an average of 50 ticket closures a week, maintaining operation reliability for staff and students
- Standardized the desktop RMA and reimaging process, reducing technician workload by 15%
- Supported conference and event spaces utilizing Ellucian for staff and students using the space
- Authored and maintained documentation for troubleshooting YubiKeys and desktop imaging

Aruba Networking - Hewlett Packard Enterprise

Roseville, CA

Software Engineering Intern

Jun 2024 - Aug 2024

- Collaborated with six others to implement new features while improving stability of Aruba switches
- Optimized a Python-based debugging utility, reducing its memory footprint by 31% for enhanced scalability
- Accelerated network debugging by implementing a C-based utility to streamline VLAN and VRF data retrieval
- Verified the reliability of AMD PSM and DSS features through creating BGP and OSPF tests

PROJECTS

TSK 200 (Github)

github.com/Zarcix/TSK200

Dec 2024 - Present

- Implement a multiprocessing instruction set utilizing semaphores and interprocess communication
- Utilized interprocess communication to have TSK nodes communicate
- Designed an execution model for nodes to execute instructions concurrently while remaining thread safe
- Debugged complex threading issues with GDB, improving stability
- **Skills Involved:** C, GDB, Linux SysCalls, Interprocess Communication, Threaded Execution, Semaphores

Noisrucer (Web Backend)

scholarlycommons.pacific.edu/rcs/2024/events/4/

Jan 2024 - May 2024

- Continued work in a team of two on the remote database used for saving game actions for analysis
- Restructured a remote database backend, enhancing data organization and retrieval efficiency
- Debugged REST calls created on the game side of the research project to find issues with log generation
- Implemented JEST-based testing, ensuring backend stability under extreme conditions
- **Skills Involved:** Javascript, Github, NPM, Backend Development, REST

SpeedReed (Github)

github.com/Zarcix/TSK200

Dec 2023 - Jan 2024

- Developed a lightweight reading application to aid language learners by displaying words based on WPM
- Created a lightweight application to display text one word at a time given a specific WPM
- Engineered a parsing system compatible with multiple languages
- **Skills Involved:** Rust, Language Parsing, Github, Development Framework

NetJoystick (Github)

github.com/Zarcix/NetJoystick

Mar 2023 - Aug 2023

- Architected a server and client model to send Joystick input over UDP to another device
- Conducted API stability assessments for Linux kernel integration
- Implemented a multi-threaded application for both the backend and frontend
- **Skills Involved:** Rust, Multi Threading, API, Frontend and Backend communication, UDP Packets, Linux Kernel Interaction, HCI

Noisrucer (Unity)

scholarlycommons.pacific.edu/rcs/2024/events/4/

Aug 2022 - May 2023

- Collaborated in a team of three on a research project to teach new computer science students recursion
- Took charge by implementing a recursive subcommand tree for structured game logic
- Added tests to ensure command tree structures were valid, including tests for backtracking and conditional checking
- Developed a backend logging system for comprehensive game activity analysis
- **Skills Involved:** C#, Github, Subscriber Publisher Model, Unity

Spork (Github)

github.com/Zarcix/Sturdy-Spork

Sep 2022 - Jan 2023

- Measured mobile application's data through reverse engineering
- Developed a backend logging system for comprehensive game activity analysis
- Created a reverence engineered version of an android application for use on all major Operating Systems
- Programmed application to be able to stream video at a high rate using TCP
- **Skills Involved:** Rust, HTTP Requests, HTTP Returns, API Usage, Testing and Debugging, Reverse Engineering

Segfault (Github)

github.com/Zarcix/Segfault

Mar 2022 - May 2022

- Led a team of three to develop a Java-based ACM Graphics game
- Ensured smooth asynchronous code integration via GitHub
- Managed project sprint planning and assigned tasks to meet development deadlines
- **Skills Involved:** Github, Java, Documentation Analysis, OOP, Project Management

Dalamalanguage (Github)

github.com/Zarcix/dalamalanguage

Jan 2022 - May 2022

- Co-developed a new programming language, focusing on variable parsing and mathematical operations
- Implemented looping support and structured grammar for functional execution
- Designed an error-handling system for scanning, parsing, and evaluation stages
- **Skills Involved:** Rust, Programming Language Development, Context-Free Grammar

TECHNICAL SKILLS

- Programming Languages:
 - Advanced: Rust, Python, Java, C++, C#, C, Javascript
 - Proficient: Bash, MIPS Assembly, Haskell
- Operating Systems: Linux (Arch, Debian, Fedora, Kali), Microsoft Windows, macOS, Android
- Networking: Cisco Switch Configuration, LAN, WAP, UDP & TCP Socket Programming
- Tools and Frameworks: Git, Gerrit, Jira, Active Directory, Docker, Kubernetes, ReactJS, Tauri
- Reverse Engineering: Wireshark, GDB, Proxy Servers
- Workflows: Git, Gerrit, Jira, Active Directory
- Spoken Languages: English (Fluent), Vietnamese (Fluent)